

Trust Ecosystems in AI Governance:

Building the Infrastructure Beneath the Rules

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Abstract

Every significant AI governance framework produced in the last five years — the EU AI Act, the NIST AI Risk Management Framework, ISO/IEC 42001, the G7 Hiroshima Process principles — calls for "trustworthy AI." None of them specifies what trust infrastructure is supposed to make that trustworthiness real. This paper argues that this omission is not a drafting detail left to implementers. It is the central problem in AI governance, and it is actively causing harm. We develop a working theory of the trust ecosystem as the appropriate unit of analysis for AI governance: a structured arrangement comprising attestation infrastructure, verification mechanisms, shared interpretive frameworks, and accountability channels for when trust fails. Against this framework, we assess where current instruments succeed, where they fall short, and why those gaps are not accidental — they are the product of identifiable incentive problems, sovereignty concerns, and coordination failures that rules alone cannot resolve. The paper then asks what a functional trust ecosystem would actually look like for AI systems operating across the EU-US-UK triangle, drawing on analogies from financial clearing systems and ICAO's aviation safety regime while being honest

about where those analogies break. We close with four specific, actionable recommendations directed at regulators, standards bodies, and AI developers. The core argument is that trust is not a property of AI systems — it is a property of the ecosystems within which those systems are embedded. Governance that treats trustworthiness as an engineering problem will keep failing. The work ahead is institutional, relational, and infrastructural.

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1. The Governance Gap That Isn't Being Named

1.1 What Frameworks Say

The past five years have produced an unusually dense body of AI governance instruments. The European Union's AI Act — the most legally ambitious attempt to regulate artificial intelligence at scale — entered into force in August 2024. The NIST AI Risk Management Framework, now in its first full version, offers a voluntary but widely referenced architecture for thinking about AI risk. ISO/IEC 42001 establishes a management system standard for AI that follows the familiar Annex SL structure recognizable to any organization already operating under ISO 9001 or 27001. The G7 Hiroshima Process produced eleven guiding principles for advanced AI that have been adopted by reference across multiple national policy documents. The OECD AI Principles underpin all of the above, and the Council of Europe AI Convention — the first binding multilateral treaty on the subject — adds a further layer for its signatories.

What these instruments share, beyond their breadth and ambition, is a common vocabulary. They all use the phrase "trustworthy AI," or some close cognate: "reliable," "safe," "human-centric," "responsible." The NIST RMF structures its entire architecture around the

concept of trustworthiness, decomposing it into characteristics including validity, reliability, explainability, privacy, and security. The EU AI Act's preamble invokes trust as a precondition for beneficial AI adoption. The Hiroshima Process principles make trustworthiness their organizing theme.

None of these frameworks specifies, with any operational precision, what infrastructure is supposed to generate the trustworthiness they call for.

This is not a minor omission. These documents are not silent on procedure — the EU AI Act runs to hundreds of articles and recitals, specifying conformity assessment procedures, technical documentation requirements, registration obligations, and enforcement mechanisms in considerable detail. The NIST RMF provides elaborate guidance on governance, mapping, measurement, and management. ISO 42001 imposes a management system architecture with defined roles, policies, objectives, and internal audit requirements. The instruments are not thin. They are, in their own terms, thorough.

And yet: when we ask what actually happens — institutionally, technically, relationally — when an AI system deployed by a hospital in Lyon interacts with a care pathway designed by an NHS trust in Manchester, or when a hiring algorithm built by a US-based vendor is used by a German employer, or when a financial risk model trained on American data is deployed in a Singaporean bank — none of these frameworks provides a complete answer. They tell us what the developer must document, what the deployer must assess, what the regulator may require. They do not tell us how trust is supposed to travel across those organizational and jurisdictional boundaries, and what would need to be true of the infrastructure for it to do so reliably.

That is the governance gap we are identifying. It is not a gap in the rules. It is a gap beneath the rules.

1.2 Why Omission Is Not Benign

The Core Problem

Current AI governance instruments mandate trustworthiness as a property of AI systems, assessed at the point of development or deployment, by the entity responsible for that system. They do not create or specify the *infrastructure* through which trust claims can be transmitted, verified, contested, and updated across organizational and jurisdictional boundaries. This is not a technical detail. It is the central architectural deficit in the current governance landscape. The consequence is that "trustworthy AI" remains a compliance declaration rather than a verifiable condition — and that the absence of verifiability is itself a source of harm.

The claim that this omission is causing harm in 2026 is not speculative. It is visible in several specific failure patterns that have become apparent as major AI deployments have moved through their first years of operation under existing frameworks.

Consider the situation of public healthcare AI across EU member states. Several member states have deployed AI-assisted diagnostic tools in hospital settings — tools that were assessed and

certified under their own national conformity assessment procedures prior to the AI Act's full applicability provisions entering into force. When those same tools are used in cross-border contexts — for a patient treated in one member state by a specialist located in another, or for a trial that draws data from institutions in multiple jurisdictions — no shared infrastructure exists to extend the trust assessment across the boundary. Each deployment is treated as a new compliance question. The original conformity assessment does not travel. The result is either regulatory paralysis (each cross-border use triggers a fresh assessment) or regulatory arbitrage (cross-border use proceeds without any assessment, exploiting the gap between the jurisdictions' authorities). Neither outcome is what the regulatory framework intended, and the harm — delayed care, suppressed innovation, or unassessed risk — is real.

Consider the supply chain transparency problem. Large language models deployed in commercial and professional settings in 2025 and 2026 are not single-vendor products. They are assembled from base models, fine-tuned by one or more parties, accessed through API layers maintained by others, and integrated into downstream applications by further developers, some of whom are small organizations with limited compliance infrastructure. The EU AI Act's obligations fall at different points in this chain — on the provider, the deployer, the importer — but the information needed to meet those obligations (about training data, algorithmic choices, performance characteristics, version histories) is not systematically transmitted through the chain. A downstream deployer has no reliable way to verify the claims made by an upstream provider, because no standardized format for those claims exists and no independent verification mechanism operates between them.

Consider finally the incident learning problem. Aviation safety works — to the extent that it does — not primarily because of pre-deployment certification, but because of a post-incident reporting and learning system that aggregates near-misses and incidents across thousands of operators globally, identifies systemic patterns, and generates mandatory safety actions. Nothing equivalent exists for AI systems. When an AI-assisted system produces a consequential error — a misdiagnosis, a discriminatory hiring decision, a flawed financial recommendation — the incident is typically handled within the organization, sometimes reported to a national regulator, and almost never shared in a form that allows cross-operator or cross-jurisdictional learning. The same failure modes are being independently discovered, and independently producing harm, at institutions that have no mechanism for informing each other.

These are not edge cases. They are the predictable consequence of building a governance architecture that specifies rules at the point of deployment but does not build the infrastructure for trust to function continuously across the systems those rules are supposed to govern.

2. What Trust Actually Means in Sociotechnical Systems

2.1 Trust vs. Confidence vs. Compliance

Key Distinction

Trust is an attitude held by one party toward another under conditions of vulnerability and incomplete information. It involves expectations about the other party's competence, integrity, and goodwill — and a willingness to be exposed to the risk that those expectations prove wrong.

Confidence is expectation without vulnerability — the expectation that a system will function as designed, based on prior experience or engineering assurance, without a relational dimension.

Compliance is conformance with specified requirements, assessed against defined criteria, without necessarily generating either trust or confidence in the underlying system's behavior outside those criteria.

The distinction matters enormously for AI governance. The philosopher Annette Baier, in her foundational 1986 analysis, identified trust as a three-place relation: A trusts B with C — where C is some good, value, or aspect of A's wellbeing that A entrusts to B. Trust is not binary. It is specific, contextual, and relational. Philip Pettit's work on the conditions for rational trust adds a further dimension: trust is warranted, on his account, when the trusted party has adequate incentives (reputational, legal, or relational) to honor the trust placed in them, and when those incentives are visible to the trusting party.

Current AI governance frameworks primarily generate compliance, in the technical sense above. A system that has passed a conformity assessment has been evaluated against specific criteria by a specified process. It has not thereby established a trust relationship with any of the parties who will rely on it. The conformity assessment tells us that the system met the defined criteria at the time of assessment. It tells us little about how the system will behave under novel conditions, how it will be updated, who is accountable if it fails, and whether the party relying on it has any mechanism for detecting that it has changed or failed.

This is not a criticism of conformity assessment as a tool. It is an observation about what compliance instruments can and cannot do. They generate a snapshot of conformity. They do not generate the ongoing relational and institutional structure that trust requires. The gap between the two is what trust infrastructure is designed to fill.

2.2 The Distributed Trustee Problem

In classical trust relationships — between a patient and a physician, between a client and a professional — the trustee is identifiable. There is a person or institution that holds the trusted good, whose competence and integrity can be assessed, and who bears accountability if the trust is betrayed.

AI systems complicate this structure in ways that are not yet fully reflected in governance design. When an AI system makes a consequential decision or recommendation, who is the trustee? The entity that trained the base model holds some relevant knowledge, but may have no contractual relationship with the end user. The entity that fine-tuned the model made choices

that shaped its behavior, but may have done so without full knowledge of the deployment context. The entity that deployed the system exercised choices about integration, configuration, and monitoring, but may have relied on assurances from upstream parties that are difficult to verify. The user bears the consequence, but typically has no visibility into any of the above.

This is what we call the distributed trustee problem: the relevant accountability is distributed across a supply chain in ways that make it very difficult for any single party — including the regulator — to hold the whole picture. The EU AI Act attempts to address this through its chain-of-responsibility provisions, which assign different obligations to providers, deployers, importers, and distributors. But assigning obligations is not the same as creating infrastructure for trust. The obligations only generate trust if the information needed to verify them is transmitted through the chain in a verifiable, standardized form — and that infrastructure does not currently exist.

2.3 Layers of Trust

We can usefully distinguish three layers at which trust operates in AI governance contexts, each with its own requirements and failure modes.

The first layer is **user-to-system trust**: the relationship between an individual or organization and an AI system they rely on. This layer has received the most attention in governance design — explainability requirements, transparency obligations, and human oversight provisions all address it. But user-to-system trust cannot exist in isolation. A user cannot assess the trustworthiness of an AI system directly; they can only assess the claims made about it and the institutional structures that stand behind those claims.

The second layer is **institution-to-institution trust**: the relationships between the organizations involved in developing, deploying, auditing, and overseeing AI systems. This layer has received less attention, but it is in some respects the load-bearing layer. Trust between institutions — expressed through contracts, audit relationships, certification schemes, and regulatory enforcement — is what allows the overall system to function at a scale that individual trust relationships cannot reach. The absence of standardized trust transmission mechanisms between institutions is the most immediate practical deficit in the current governance landscape.

The third layer is **cross-border trust propagation**: the conditions under which trust established in one jurisdiction can be recognized or extended in another. This layer is the least developed and the most consequential for AI systems that operate, as most significant systems do, across multiple jurisdictions simultaneously. The EU's adequacy decisions in data protection, and mutual recognition agreements in product safety, represent partial models for cross-border trust propagation in adjacent domains. AI governance has not yet developed equivalent mechanisms.

3. Anatomy of a Trust Ecosystem

Defining the Trust Ecosystem

A **trust ecosystem** is a structured arrangement of institutions, standards, infrastructure, and practices that enables trust claims to be generated, transmitted, verified, contested, and updated across organizational and jurisdictional boundaries. It comprises four components:

1. **Attestation Infrastructure:** The mechanisms by which trust claims are formally made and recorded — who says what about which system, in what form, with what standing.
2. **Verification Mechanisms:** The processes and infrastructure by which attestation claims can be independently checked — including technical audit tools, third-party evaluation bodies, and ongoing monitoring systems.
3. **Shared Interpretive Frameworks:** The common vocabularies, schemas, and standards that allow trust claims made in one context to be understood and evaluated in another — without which verification cannot occur across organizational or jurisdictional boundaries.
4. **Accountability and Failure Channels:** The mechanisms by which trust failures are detected, reported, responded to, and learned from — including incident reporting systems, liability frameworks, and redress mechanisms. A trust ecosystem without functional failure channels is not a trust ecosystem. It is a certification theater.

3.1 Attestation Infrastructure

Attestation is the act of formally asserting that something is the case — that a system has certain properties, was built in a certain way, was tested against certain criteria, is operated within certain constraints. Attestation is the entry point into the trust ecosystem: without it, there is nothing for verification mechanisms to check, no claims for interpretive frameworks to make legible, and no basis for accountability channels to act on.

Current AI governance instruments generate attestation in several forms. SOC 2 reports — originally designed for cloud service providers — are widely used by AI developers to attest to security and availability properties. EU AI Act conformity assessments require technical documentation that, for high-risk AI systems, must be reviewed by a notified body. ISO 42001 certification requires an external audit against the management system standard. Model cards and datasheets, while not legally mandated in most jurisdictions, have become a de facto attestation format for many AI developers, particularly in the research community.

The problem is not that attestation instruments do not exist. It is that they are fragmented, non-standardized, not machine-readable, not systematically linked to the systems they describe, and not maintained across system lifecycles. A SOC 2 report covers a point in time. A conformity assessment addresses the system as submitted for assessment. An ISO 42001 certificate speaks to the management system, not to any specific AI output. Model cards, when they exist at all, are not verified by any external party and are not updated when the model changes.

Functional attestation infrastructure requires more than the existence of attestation documents. It requires: recognized attestation bodies with defined competence requirements; standardized claim formats that allow claims to be compared and aggregated; liability frameworks that give attestors an incentive to attest accurately; and machine-readable formats that allow attestation data to flow through systems that rely on it. None of these conditions is currently met at scale in any jurisdiction.

What functional attestation infrastructure would require, in concrete terms, is a set of agreed claim types — covering provenance, capability characteristics, testing methodology, deployment constraints, update history, and known limitations — expressed in structured, machine-readable formats, made by identifiable parties with defined liability exposure, and linked persistently to the specific system version to which they refer. The analogy from financial services is the Legal Entity Identifier (LEI) system: a globally harmonized identifier for legal entities involved in financial transactions, maintained by a network of accredited Local Operating Units, that makes it possible to trace counterparty relationships across borders and systems. AI governance needs something equivalent for system identity — and then a further layer for the claims made about those systems.

As a micro-scale demonstration of what functional attestation infrastructure looks like in practice, we note HHI's own implementation. Provenance manifests ([provenance.json](#)) and authority records ([authority.json](#)) are maintained for each publication in the Research_Papers repository, establishing machine-readable authorship lineage. AI co-creation disclosure files ([model-use.md](#)) document the co-creation process in structured form. Governed authorship README sections provide human-readable attestation within the repository surface. Lineage manifests index the co-creation evidence trail across publication versions. These artifacts are not supplementary to HHI's governance infrastructure — they are its attestation layer, providing verifiable, machine-readable evidence of authorship provenance for every institutional output. The architecture is simple, open, and replicable. Its existence is evidence that attestation infrastructure of the kind described in this section does not require centralized registries or proprietary platforms to begin functioning — it requires institutional commitment and structured artifact discipline.

3.2 Verification Mechanisms

Attestation without verification is declaration. For trust claims to be operationally meaningful — to actually support the reliance decisions of downstream parties — they must be checkable. Verification mechanisms are the processes and infrastructure by which an independent party, or the relying party itself, can assess whether an attestation claim is accurate.

The challenge of verification in AI systems is substantially more difficult than in most prior domains of product regulation. AI systems are not static. A pharmaceutical product, once approved, has a defined composition that can be tested against its specification. An aircraft component can be physically inspected. An AI system, by contrast, can change — through fine-tuning, through context modification, through retrieval-augmented generation that alters effective behavior without changing model weights — in ways that may not be visible from outside the system and may not trigger any re-assessment obligation under current frameworks.

Model opacity compounds the verification problem. The internal representations that drive AI system behavior are not, in the general case, humanly interpretable. Current explainability techniques provide local approximations of model behavior in specific instances, not auditable accounts of the rules or representations underlying that behavior. This means that verification, in the AI context, cannot rely on the inspection of internal mechanisms in the way that software audit can inspect code. It must rely instead on behavioral testing — evaluating system outputs across defined test sets — which raises its own questions about test coverage, adversarial robustness, and the gap between tested and deployed behavior.

Data provenance adds a further layer of difficulty. The properties of an AI system are substantially determined by its training data — its coverage, its biases, its temporal scope, its labeling quality. Verification of training data claims is extremely difficult without direct access to the data, which creates a fundamental tension with the legitimate commercial and privacy interests of AI developers.

Current instruments are almost entirely reactive in their approach to verification. Conformity assessments happen before deployment. Post-deployment monitoring obligations are vague. Incident reporting is, where it exists, focused on specific adverse events rather than on systematic drift or degradation in system properties. A proactive, continuous verification infrastructure would require: technical tools for behavioral monitoring at scale; organizational capacity to conduct ongoing audits; information-sharing mechanisms that allow monitoring data to be aggregated across deployments; and regulatory authority to require access to the information needed for meaningful verification. None of this exists in functional form.

HHI's verification infrastructure illustrates what the lowest layer of this component looks like when operationalized. SHA256 manifests ([SHA256SUMS](#)) and canonical checksum inventories bind governance artifacts to their content at the time of generation. Integrity manifests document artifact integrity across the ecosystem. Checksum enforcement workflows make verification a routine step in governance operations rather than an ad hoc forensic exercise. The [governance_audit_exports/](#) bundles are structured for external verification: a reviewer with access to the [SHA256SUMS](#) and the [MASTER_EVIDENCE_INDEX.log](#) can verify artifact integrity without internal access to HHI's systems. This is not a claim that HHI's verification infrastructure meets the requirements of a regulated trust ecosystem. It is a demonstration that the underlying technical primitives — checksum binding, structured export, index-linked evidence — are available, functional, and not technically demanding to deploy.

3.3 Shared Interpretive Frameworks

Even where attestation is made and verification is technically possible, the practical operation of a trust ecosystem requires that the claims made in one context can be understood and evaluated in another. This is the function of shared interpretive frameworks: the common vocabularies, schemas, taxonomies, and standards that make trust claims legible across organizational and jurisdictional differences.

The absence of shared interpretive frameworks is most visibly felt at cross-jurisdictional boundaries. The EU AI Act classifies AI systems by risk level using criteria that do not map

cleanly onto the risk categorizations used by NIST, by UK regulators, or by the sector-specific frameworks — such as the FDA's approach to AI-enabled medical devices — that operate in parallel. An AI developer seeking to demonstrate compliance across multiple jurisdictions faces the prospect of generating entirely separate documentation sets, each structured around the interpretive categories of a different framework. A regulator in one jurisdiction seeking to evaluate a system assessed under another jurisdiction's framework has no common reference point for interpreting the assessment.

Shared interpretive frameworks are also necessary within jurisdictions, across the different AI system types and deployment contexts that a single framework must cover. "Risk" means something different for a hiring algorithm, a medical diagnostic tool, a financial recommendation system, and a content moderation system — different in terms of the relevant failure modes, the appropriate performance metrics, the baseline against which improvement is measured, and the affected population whose interests must be considered. A framework that does not specify how these contextual differences should be reflected in attestation and verification claims cannot generate trust claims that are meaningful across contexts.

The development of shared interpretive frameworks is, in our assessment, the component of trust ecosystem infrastructure where the greatest current effort is being applied — through standards bodies, through the development of common vocabularies in academic and civil society settings, and through regulatory dialogue processes. ISO/IEC JTC 1/SC 42 has produced a growing body of AI standards that provide vocabulary and conceptual infrastructure. The NIST AI RMF provides a risk management framework that is beginning to develop cross-reference mappings to other instruments. These are genuine contributions. But they fall short of functional shared interpretive frameworks in a key respect: they provide common vocabulary without specifying how claims expressed in that vocabulary should be validated, and without creating the institutional infrastructure for interpreting claims made in different frameworks as equivalent for trust purposes.

3.4 Accountability and Failure Channels

The final component of a functional trust ecosystem is perhaps the most consequential and the most neglected: the mechanisms by which trust failures are detected, reported, responded to, and learned from. No attestation infrastructure will be perfect. No verification mechanism will catch every drift or degradation. No interpretive framework will anticipate every edge case. The resilience of a trust ecosystem depends, in the end, on what happens when it fails.

ICAO's occurrence reporting system provides the most instructive available model. Under ICAO Annex 13 and the complementary framework of the Safety Management Manual, aviation accidents and serious incidents must be reported to national authorities and to ICAO's accident database. Beyond mandatory reporting, the ICAO framework encourages voluntary reporting of near-misses and safety concerns through protected reporting systems that provide legal immunity to reporters, ensuring that the incentive to report is not undermined by fear of enforcement consequences. The resulting data is analyzed for systemic patterns, and the conclusions are published as safety recommendations — binding on state parties — that apply across the global aviation system.

The result is a closed learning loop: incidents generate data; data generates analysis; analysis generates recommendations; recommendations generate changes in practice and design; and the cycle repeats. The safety improvement that has made commercial aviation the safest form of mass transport in human history is not principally the product of pre-flight certification — it is the product of this continuous learning system operating over decades.

AI governance has no equivalent. When an AI system fails consequentially — and in 2025 and 2026, consequential AI failures have become a regular feature of the governance landscape — the failure is typically handled within the organization, occasionally reported to a sectoral regulator, and almost never shared in a form that supports cross-operator learning. The same failure modes are being independently discovered by institutions that have no mechanism for informing each other, and the aggregate harm is multiplied accordingly.

Building functional accountability and failure channels requires several things: a mandatory incident reporting framework with defined scope and thresholds; protected reporting mechanisms that reduce the disincentive created by liability exposure; a cross-border data-sharing architecture that allows incident data to be aggregated and analyzed across jurisdictions; and an institutional actor with the mandate and capacity to conduct and publish systemic analysis. This is not a simple infrastructure to build, and it raises genuine questions about liability, sovereignty, and competitive disclosure. But the aviation precedent demonstrates that it is achievable — and that the alternative, a governance framework without a learning loop, is not actually providing the safety assurance it claims to provide.

4. The Propagation Problem

4.1 How Trust Travels

Trust does not travel automatically. The fact that a trust relationship has been established in one context does not mean that it can be relied on in a different context, by a different party, across a different organizational or jurisdictional boundary. Trust propagation — the extension of trust established in one context to another — requires specific institutional mechanisms, and those mechanisms are precisely what AI governance currently lacks.

We can identify three models from adjacent domains that illustrate how trust propagation has been achieved in other high-stakes fields.

The ICAO safety regime represents the most ambitious multilateral trust propagation architecture in any technical domain. States that ratify the Chicago Convention commit to implementing Standards and Recommended Practices (SARPs) that have been adopted by ICAO's governing bodies. An aircraft certified as airworthy in its state of manufacture is recognized as airworthy by every other contracting state that has implemented the relevant SARPs — not because each state independently verifies the certification, but because the shared framework generates mutual recognition of the certifying authority's competence. The

trust travels with the certificate, because the certificate was issued under a framework that the receiving state has agreed to treat as equivalent to its own.

Pharmaceutical mutual recognition agreements represent a bilateral or regional variant of the same logic. The EU's mutual recognition procedures allow a marketing authorization granted by one member state to be recognized by others on the basis of a shared assessment process and shared standards. EU-US mutual recognition agreements for pharmaceutical inspections have reduced duplication of facility inspections by allowing each party to rely on the other's inspection findings — but only for specified facility types, after a confidence-building process that itself took years to complete.

The Basel Accords provide a third model: internationally agreed standards for bank capital adequacy that are not themselves legally binding, but that have been incorporated into the domestic law of over 100 jurisdictions, creating de facto harmonization that allows trust to propagate through the financial system even in the absence of a formal multilateral treaty. The standards were developed by a committee of central banks and banking supervisors operating under G10 auspices, giving them sufficient regulatory authority to serve as a coordination focal point without requiring formal international legal status.

None of these models maps cleanly onto the AI governance context. ICAO works in part because aircraft can only be operated by licensed pilots in certified aircraft under supervised conditions — the technical characteristics of aviation create natural choke points for regulatory control that AI systems do not have. Pharmaceutical mutual recognition works because the subject matter — drug compounds — is physical, testable, and relatively stable. Basel works because banking regulation operates through a small number of licensed, heavily supervised institutions — not through millions of developers and deployers with widely varying compliance capacity. These differences are real and we do not minimize them. But the models demonstrate that trust propagation is achievable across international boundaries, and they illuminate what structural features make it possible: shared standards, recognized assessment bodies, mutual commitments, and institutional actors with the mandate to manage the system.

4.2 Technical Barriers to Trust Propagation

The technical barriers to AI trust propagation are substantial and largely unresolved. Three merit particular attention.

The first is model opacity, discussed above in the context of verification. For trust to travel, the party receiving a trust claim must have some basis for evaluating it. In domains where the subject matter is physically inspectable or mathematically tractable, this basis can be provided by shared testing protocols — both parties apply the same tests and compare results. For AI systems, the lack of interpretable internal mechanisms means that behavioral testing is the primary available tool, and behavioral testing of systems with very large output spaces cannot provide the same level of assurance as structural inspection.

The second is data provenance. An AI system's behavior is substantially determined by its training data. Two systems with identical architectures trained on different data may behave very differently in deployment. For trust to travel across jurisdictions, the receiving party needs to be

able to assess not just what the system does on a standard test set, but what data shaped its behavior — and whether that data is appropriate for the deployment context in the receiving jurisdiction. Current frameworks do not require training data disclosure in a form that would enable this assessment, and the commercial sensitivity of training datasets creates strong resistance to such requirements.

The third is runtime divergence. AI systems can behave differently in deployment than they did in testing — through distribution shift (the deployment environment differs from the training environment), through adversarial inputs (users deliberately attempting to elicit atypical behavior), or through dynamic components (retrieval-augmented systems whose effective behavior is partly determined by external data sources that may change). A trust claim established at the time of assessment may not accurately characterize the system's behavior at the time of reliance, and no technical mechanism currently exists to maintain continuous alignment between assessed and deployed behavior.

4.3 Institutional Barriers to Trust Propagation

The institutional barriers to trust propagation are, if anything, more intractable than the technical ones. We identify three principal structural forces.

The first is liability asymmetry. When trust is propagated across an organizational boundary — when party A relies on an attestation made by party B about a system developed by party C — the question of who bears liability for a failure of that trust is not answered by any existing legal framework in a way that supports voluntary trust propagation. The party that makes an attestation has an incentive to make it conservatively — to understate capability, overstate limitation, and resist being held responsible for downstream uses of the claim. This incentive structure systematically degrades the quality and completeness of attestation data, because the parties best placed to provide it have the strongest reasons to provide it minimally.

The second is competitive disclosure aversion. AI capabilities are, in the current commercial landscape, a source of significant competitive advantage. The information that would be most useful for trust propagation — detailed training data characteristics, fine-tuning methodologies, red-teaming results, known failure modes — is also the information that AI developers most strongly wish to protect. Mandatory disclosure regimes have addressed this tension in other domains (clinical trial disclosure, financial reporting) through a combination of legal compulsion and safe-harbor provisions. No equivalent mechanism exists for AI systems.

The third is regulatory divergence. The EU AI Act, the NIST RMF, the UK's sector-led approach, and the various national frameworks emerging in Asia and the Global South reflect genuinely different regulatory philosophies, institutional arrangements, and political commitments. The EU Act is a binding legal instrument with enforcement authority; the NIST RMF is voluntary guidance. The EU categorizes AI systems by risk class with defined legal consequences; the UK's approach is sector-specific and principle-based. These differences are not simply matters of drafting — they reflect different views about the appropriate role of government in technology markets, different institutional capacities for enforcement, and different political economies of AI development. Trust propagation across these divergent frameworks requires either convergence

(which is politically difficult) or mutual recognition (which requires the confidence-building process that we explore in Section 6).

4.4 Case Studies in Trust Propagation Failure

Two case studies illustrate the propagation problem in concrete terms.

Healthcare AI across EU member states. AI-assisted diagnostic and decision-support tools have been deployed in clinical settings across the EU since before the AI Act's full provisions entered into force. Some of these tools were assessed and approved under national frameworks — MDR (Medical Device Regulation) or national AI regulatory guidance — before the AI Act's classification system became applicable. In cross-border clinical research and telemedicine contexts, these tools are used by practitioners in member states that had no part in the original assessment. No mechanism exists for the original assessment to be extended or recognized across the border. The treating clinician has no way to verify the assessment, the deploying institution has no way to know whether the assessment remains current (given the software update cycles of most clinical AI tools), and the receiving regulator has no formal basis for recognizing the originating state's assessment as adequate. The practical result is a mosaic of informal risk-tolerance decisions by individual institutions — which is precisely the kind of unstructured governance outcome that the AI Act was designed to prevent.

Large language models across the EU-UK-US triangle. The major general-purpose AI models deployed in 2025 and 2026 are developed primarily in the United States, subject to EU AI Act obligations as providers to EU users, regulated by UK authorities under the framework of the AI Safety Institute's evaluation process, and used by institutions in all three jurisdictions that have their own downstream obligations. The developer must simultaneously meet EU technical documentation requirements, comply with UK voluntary evaluation processes, and satisfy US customer procurement requirements — each structured around different risk vocabulary, different documentation formats, and different verification expectations. The result is not harmonization but layering: separate compliance artifacts for each jurisdiction, none of which references or builds on the others, and none of which gives any party in another jurisdiction a reliable basis for relying on the assessments conducted elsewhere. Trust does not propagate through this system. It is generated independently, expensively, and redundantly, in each jurisdiction — or it is not generated at all.

5. Where Current Frameworks Fall Short

5.1 Framework Assessment Table

The following table assesses five major current AI governance instruments against the four components of the trust ecosystem framework developed in Section 3. Ratings reflect our assessment of the extent to which each instrument addresses each component in a way that contributes to functional trust infrastructure, not merely to nominal compliance requirements.

Framework	Attestation Infrastructure	Verification Mechanisms	Shared Interpretive Frameworks	Accountability & Failure Channels
EU AI Act	Partial. Requires technical documentation and conformity assessments for high-risk systems. Does not specify standardized claim formats or machine-readable requirements. Coverage limited to defined risk classes.	Partial. Notified body oversight for high-risk systems. Post-market monitoring required but vaguely specified. No continuous behavioral monitoring infrastructure.	Partial. Provides a risk classification taxonomy, but one not harmonized with other major frameworks. Standards referenced but not mandated in detail.	Weak. Serious incident reporting required for high-risk systems deployed in certain contexts. No cross-border aggregation mechanism. No protected reporting system.
NIST AI RMF	Weak. Provides a governance vocabulary but not attestation requirements. Profile framework allows organizations to document their risk management approach but does not specify external attestation.	Weak. Addresses measurement as a function but does not specify verification infrastructure or third-party evaluation requirements.	Strong. The most developed common vocabulary in the current landscape. Cross-reference mappings to other frameworks are being developed. Not legally binding, limiting its coordinating function.	Weak. Addresses govern and respond functions at conceptual level. No incident reporting requirements or cross-organizational learning mechanisms.
ISO/IEC 42001	Moderate. Provides a certifiable management system standard with external audit. Does not address	Moderate. Requires internal audit and management review. External certification audit provides some independent	Moderate. Annex SL structure provides consistency with other ISO management systems. Definitions	Weak. Requires nonconformity and corrective action procedures. Does not address external incident reporting or

	system-level attestation. Certificate covers the management system, not the AI system outputs.	verification of management processes, not system behavior.	aligned with ISO/IEC 22989. Not jurisdiction-specific.	cross-organizational learning.
G7 Hiroshima Process	Minimal. Eleven principles articulate desired properties but do not specify attestation requirements or mechanisms.	Minimal. Calls for testing and evaluation without specifying infrastructure.	Moderate. Provides a politically significant statement of shared principles that can serve as a reference point for framework harmonization. Not operationally specified.	Minimal. Voluntary principles with no enforcement mechanism or incident reporting framework.
UK Sector-Led Approach	Partial. Sector regulators (FCA, CQC, ICO, etc.) imposing attestation requirements within their domains. Not harmonized across sectors.	Partial. AI Safety Institute evaluation process for frontier models is the most developed proactive verification mechanism in any major jurisdiction. Scope limited to frontier models and voluntary participation.	Weak. Deliberate avoidance of cross-sectoral framework means shared interpretive vocabulary is not being systematically developed. Cross-regulator forum provides coordination but not standardization.	Partial. Sector-specific incident reporting in regulated sectors (financial services, healthcare). No cross-sector AI-specific reporting system.

5.2 Structural Gaps and Why They Are Not Accidental

The pattern in the table above is not random. The gaps cluster in predictable ways that reflect structural incentive problems, not drafting oversights. Understanding why the gaps exist is necessary to understanding what it would take to close them.

The collective action problem on attestation standards. Standardized, machine-readable attestation formats would benefit the overall ecosystem — they would reduce compliance costs

for organizations operating across jurisdictions, enable automated verification, and support cross-border trust propagation. But any individual actor that moves first bears the cost of developing the standard, while others can free-ride on the infrastructure once it exists. AI developers have an additional incentive against detailed attestation: disclosure of system characteristics creates competitive exposure. The result is a predictable underinvestment in attestation infrastructure by private actors, which can only be resolved by regulatory mandate or by a coordinating body with sufficient authority to create and enforce standards.

The disclosure problem on verification. Meaningful verification of AI systems requires access to training data, model weights, evaluation results, and deployment data — all of which AI developers treat as proprietary. No jurisdiction has successfully mandated this disclosure in a form that supports independent verification, because the political economy of AI development strongly disfavors disclosure requirements. The EU AI Act's technical documentation requirements come closest, but they apply only to high-risk systems and are assessed by notified bodies under confidentiality, not published for independent review. The gap between what meaningful verification requires and what disclosure regimes currently mandate is large and shows no sign of closing quickly.

The sovereignty problem on accountability channels. Cross-border incident reporting — the mechanism most needed for the kind of systemic learning that aviation safety achieves — requires each participating jurisdiction to share information about failures within its borders with a cross-border body that can analyze and disseminate it. This involves a degree of regulatory transparency that raises genuine sovereignty concerns: sharing incident data means sharing information about the AI systems operating in critical national infrastructure, about the failures of regulated entities under domestic law, and about national regulatory capacity. These concerns are not irrational, and they explain why cross-border accountability channels are the most underdeveloped component of current governance infrastructure. They also suggest that building such channels will require confidence-building processes similar to those that preceded pharmaceutical mutual recognition agreements — a timeline measured in years, not months.

The first-mover disadvantage in standards. Jurisdictions that invest in developing detailed AI governance standards create frameworks that embed their own institutional preferences and technical assumptions. Other jurisdictions face a choice between adopting a foreign framework (ceding standard-setting authority) and developing their own (creating fragmentation). This dynamic has played out in financial services, telecommunications, and pharmaceutical regulation, where dominant standards typically reflect the preferences of the jurisdiction with the largest market. In AI governance, the EU's early regulatory action and the US's market dominance create a predictable tension that is already visible in divergences between the AI Act and the NIST RMF. Resolution requires sustained multilateral engagement — not the announcement of principles, which is easy, but the detailed technical work of cross-reference mapping and mutual recognition, which is hard.

6. What Functional Trust Infrastructure Looks Like

6.1 Partial Models and Their Limits

We have drawn, throughout this paper, on analogies from ICAO's aviation safety regime and from financial clearing infrastructure. We have done this deliberately, because these analogies illuminate what structural features make trust propagation possible at scale. We owe it to the reader to be equally clear about where the analogies break.

The ICAO model works in part because aviation is genuinely global in its economics and its physics. An aircraft that cannot fly in the airspace of non-ICAO states is commercially unviable, which gives aviation actors strong incentives to accept the multilateral framework even when it imposes costs. AI systems do not have this characteristic — a system that cannot operate in a non-participating jurisdiction may simply be redirected to participating markets, reducing the incentive to accept multilateral constraints. ICAO also works because the state of manufacture has a strong reputational and legal interest in the safety of aircraft bearing its certificate — the certification authority of a state whose aircraft fail is a diplomatic and commercial problem for that state. AI governance does not yet have an equivalent accountability structure linking the developer's jurisdiction to the consequences of system failure.

Financial clearing infrastructure provides a different set of useful features. SWIFT's messaging standards — and more recently the ISO 20022 migration — demonstrate that very large numbers of institutions with conflicting interests can be brought to adopt a common technical standard when the network benefits of doing so are sufficiently large and the standard is managed by a trusted non-governmental body. The key features of SWIFT's success are worth noting: it is membership-based, with network access contingent on compliance; it is governed by a cooperative structure that aligns member interests; it provides genuine operational value (not just compliance value) to members; and it has sufficient market power to make non-participation economically costly. None of these features is present in current AI governance infrastructure — but they suggest what design choices would need to be made to achieve similar outcomes.

The limits of both analogies are also worth naming. Aviation and financial clearing operate in domains where the subject matter — aircraft safety characteristics, financial transaction records — is amenable to standardized technical description in ways that AI system behavior is not. The opacity of AI systems is not just a disclosure problem; it is an epistemic one. We do not yet have the conceptual vocabulary to describe AI system behavior in the precise, verifiable terms that ICAO SARPs use to describe aircraft performance, or that SWIFT messaging standards use to describe financial transactions. Building that vocabulary is part of the work of creating shared interpretive frameworks — and it is work that must precede, or at least accompany, the development of attestation and verification infrastructure.

6.2 A Minimal Viable Trust Ecosystem

Working from the framework developed in Section 3 and the lessons of the partial models examined in Section 6.1, we can describe what a minimal viable trust ecosystem for AI systems operating across the EU-US-UK triangle would require. We frame this as minimal viable because we are not describing the fully developed infrastructure that would be needed for

comprehensive AI governance at scale — we are describing the minimum set of components that would need to be in place for trust to begin propagating reliably across the most important current jurisdictional boundaries.

1. A federated AI system registry with interoperable identifiers. The first requirement is that AI systems can be uniquely and persistently identified across organizational and jurisdictional contexts. This requires a registry — or, more precisely, a federated network of registries with interoperable identifier formats — that assigns persistent identifiers to AI systems in a way that is tied to specific version states and maintained across updates. The analogy is the LEI system for financial entities, or the ISIN system for financial instruments: not a single global registry, but a network of accredited registrars operating under a common standard. Without persistent system identifiers, attestation claims cannot be reliably linked to the systems they describe, and verification cannot be systematically organized. This is the infrastructure precondition for everything else.

2. Shared audit trail standards. The second requirement is a common format for the structured logging of AI system governance events — decisions made about the system's development, deployment, configuration, and modification — that allows this record to be created once and read by multiple parties across organizational boundaries. The analogy is the FIX Protocol or SWIFT messaging standards in financial services: a common message format that allows different institutional systems to exchange structured information without bilateral customization. A shared audit trail standard for AI would specify what events must be recorded, in what format, with what metadata, and in what form they must be accessible to authorized verifiers. It would not need to be a single global standard from the outset — it could begin as a bilateral EU-US or EU-UK standard, as pharmaceutical mutual recognition did, with expansion as confidence builds.

3. A liability allocation framework for cross-organizational AI supply chains. The third requirement addresses the institutional barrier identified in Section 4.3: the absence of a clear liability framework for trust failures in AI supply chains. This is primarily a legal and regulatory challenge, not a technical one. It requires jurisdictions to agree — or, at minimum, for their courts and regulators to develop convergent doctrine — on how responsibility for AI system failures is allocated across the supply chain, so that attestation claims can be made by parties who have defined and bounded liability exposure. The EU AI Act's chain-of-responsibility provisions provide a starting point, but they are not sufficient on their own: they specify obligations without providing a framework for how courts should assess damages when those obligations are met but harm still occurs, and they do not address the cross-border liability questions that arise when the failure crosses jurisdictional lines.

4. A cross-border AI incident reporting network. The fourth requirement is the accountability and failure channel described in Section 3.4: a mechanism for mandatory reporting of AI incidents above defined thresholds, aggregation of that data across jurisdictions, and systemic analysis and publication of safety-relevant findings. The ICAO occurrence reporting system is the model. A bilateral start — an EU-UK agreement on AI incident data sharing, or an EU-US equivalent — would be more achievable than a global system and would demonstrate the feasibility of the model. The protected reporting provisions of aviation safety programs should be

incorporated from the outset: the experience of aviation and of nuclear safety is that voluntary reporting only functions if reporters have immunity from enforcement consequences for what they disclose.

5. Mutual recognition of conformity assessment bodies beginning with ISO/IEC 42001.

The fifth requirement is a starting point for trust propagation across jurisdictional boundaries. The most immediately achievable form is mutual recognition of the accreditation bodies that certify ISO/IEC 42001 auditors — not mutual recognition of the AI Act's notified bodies, which raises more fundamental questions about regulatory equivalence, but mutual recognition of the management system certification infrastructure that already operates under a global accreditation standard. A management system certification by an UKAS-accredited certification body would be recognized as equivalent to a certification by a DAkkS-accredited body; a certification by an ANAB-accredited body would be recognized by both. This is not, by itself, a solution to the cross-border trust propagation problem — management system certification does not speak to specific AI system properties. But it establishes the institutional infrastructure for mutual recognition, creates a precedent for regulatory cooperation, and reduces the compliance burden on organizations operating across jurisdictions.

Note on Sequencing

We present these five components as a minimal viable trust ecosystem, not as a sequential development path. In practice, the components are interdependent: a registry without shared audit trail standards is of limited value; shared standards without liability allocation create attestation without accountability; incident reporting without mutual recognition generates data that cannot be systematically acted on. The practical sequencing challenge is to identify which components can be developed in parallel and which represent genuine prerequisites for others.

Our assessment is that components 1 (registry), 2 (audit trail standards), and 5 (mutual recognition) can be developed simultaneously — they are primarily technical and institutional rather than politically contentious. Component 4 (incident reporting) requires component 5 (mutual recognition of assessment bodies) to be politically viable, because cross-border data sharing is only achievable between parties that have established a baseline of institutional trust. Component 3 (liability allocation) requires sustained legal and regulatory dialogue that should begin immediately but will not produce workable frameworks quickly. The implication is a development path in which the technical and standards-based components lead, creating the conditions for the politically sensitive components to follow.

HHI has built elements of the minimal viable trust ecosystem described in this section without waiting for regulatory mandate or industry coordination. The attestation layer is operational: `provenance.json`, `authority.json`, and `model-use.md` constitute a functioning provenance registry at institutional scale. The verification layer is operational: `SHA256SUMS` and canonical checksum inventories provide cryptographic binding. The interpretive framework layer is operational: `governance_events.jsonl` schemas provide structured, categorized event

records that create a shared interpretive surface for governance events. The accountability channel layer is operational: `stop_authority.py`, `escalation_engine.py`, and `DECISION_BOUNDARY_MAP.json` implement a tiered intervention architecture with persistent accountability records. We document this not to suggest that HHI's institutional ecosystem is equivalent to the EU-US-UK trust infrastructure this section envisions — it is not. We document it because the argument that trust infrastructure requires coordination at scale before any institution can implement it is empirically false. The infrastructure is buildable. The dependency is not technical.

7. Recommendations

We close with four specific recommendations directed at the actors best positioned to act on them. Each recommendation is framed with a defined target actor, a defined action, and a defined timeframe.

Recommendation 1 — EU AI Office and NIST: Develop a Joint Cross-Reference Framework (Target: 18 months)

The EU AI Office and NIST should commission a joint technical working group to produce a published cross-reference mapping between the EU AI Act's risk classification and documentation requirements and the NIST AI RMF's Govern, Map, Measure, and Manage functions. This mapping should be technically specific — not a statement of alignment at the level of principles, but a clause-level reference that allows an organization operating under both frameworks to understand which documentation produced for one framework satisfies which requirements of the other, where genuine differences exist, and what additional steps are needed to meet both. The EU-US Trade and Technology Council provides an existing institutional mechanism for this kind of bilateral technical work. The target timeframe of 18 months is ambitious but achievable given the maturity of both frameworks. A published cross-reference mapping would immediately reduce compliance costs for transatlantic AI developers and would establish a precedent and methodology for extending the mapping to other jurisdictions.

Recommendation 2 — ISO/IEC JTC 1/SC 42 and NIST: Publish an Open Technical Standard for AI System Audit Logs (Target: 12 months)

ISO/IEC JTC 1/SC 42, in collaboration with NIST, should publish a technical specification — as a fast-track ISO standard or as a NIST Special Publication available for international adoption — defining a common format for AI system governance audit logs. The specification should cover: the event types that must be logged (system creation, version update, training data change, configuration modification, deployment decision, incident, corrective action); the minimum metadata required for each event type; the format requirements for machine readability and

long-term archival; and the access control framework that determines which parties have the right to read, write, and audit the log. The standard should be open — available without licensing fees and implementable without proprietary infrastructure — in order to maximize adoption by small and medium-sized AI developers who lack the resources for proprietary compliance tooling. A 12-month target is achievable through a fast-track process if there is political will from the participating national standards bodies to prioritize it.

Recommendation 3 — EU AI Office and UK AI Safety Institute: Establish a Bilateral AI Incident Reporting Agreement (Target: End 2026)

The EU AI Office and the UK AI Safety Institute should negotiate and publish, before the end of 2026, a bilateral agreement establishing a shared AI incident reporting framework. The agreement should specify: the categories of AI incident subject to mandatory reporting under the framework; the reporting thresholds (by system risk class and harm severity); the data format and sharing protocol for incident reports; the confidentiality protections for reported information; the institutional arrangements for joint analysis and publication of systemic findings; and the protected reporting provisions that provide legal immunity for good-faith disclosure. The EU-UK context is the most achievable starting point because the two parties share a legal tradition, a common set of major AI deployers, and an established track record of post-Brexit regulatory cooperation in financial services. An agreement concluded before the end of 2026 would establish a model that can be extended to include the United States, and eventually other jurisdictions, in subsequent phases.

Recommendation 4 — AI Developers: Adopt Supply-Chain Trust Documentation as Standard Practice

AI developers — including base model providers, fine-tuners, and application deployers — should adopt a standard practice of publishing structured supply-chain trust documentation for AI systems deployed in consequential contexts. This documentation should include: a persistent system identifier linked to the specific deployed version; a provenance record specifying upstream components, training data sources, and fine-tuning decisions; a disclosure of known limitations and failure modes identified in pre-deployment evaluation; a commitment to version-specific update notification to downstream deployers; and a point of contact for downstream verification requests. This recommendation does not wait on regulatory mandate — the information asymmetries in current AI supply chains are causing harm now, and developers that adopt this practice will differentiate themselves in procurement contexts where deployers are increasingly demanding supply-chain transparency. We recommend that developers adopt the audit log format specified in Recommendation 2, once available, as the standard for this documentation. In the interim, machine-readable formats compatible with model card and datasheet conventions should be used.

8. Conclusion

The central argument of this paper can be stated plainly. Trust is not a property of AI systems. It is a property of the ecosystems within which those systems are embedded. When we say that an AI system is trustworthy, we mean — if we are being precise — that the institutional and technical arrangements surrounding that system are such that the claims made about it can be verified, that failures will be detected and acted on, that accountability is allocated clearly enough to create genuine incentives for responsible behavior, and that the information needed to make reliance decisions is available to the parties who need to make them. None of this is a property of the system itself. All of it is a property of the ecosystem.

The major AI governance frameworks produced in the last five years have made genuine contributions. The EU AI Act has established a legally enforceable classification and compliance system that will, over time, drive real changes in how high-risk AI systems are developed and deployed in the world's largest regulatory market. The NIST AI RMF has provided the most useful common vocabulary for AI risk management currently available. ISO/IEC 42001 has given organizations a certifiable management system framework they can adopt within existing compliance infrastructure. The G7 Hiroshima Process has provided political cover for regulatory cooperation at the highest level. We credit these contributions, and we do not argue for their replacement.

What is still missing is the infrastructure beneath the rules. The attestation layer that allows trust claims to be made in standardized, verifiable form. The verification layer that allows those claims to be independently checked. The interpretive layer that allows claims made in one jurisdictional context to be evaluated in another. The accountability layer that detects failures, aggregates learning, and creates consequences for breach. Without these components, the governance frameworks currently in operation are — in our assessment — generating compliance but not trust. The distinction matters because compliance is assessed at a point in time, while trust must be maintained continuously. Compliance is assessed against defined criteria, while trust must function across novel situations that no assessment framework fully anticipates. Compliance is demonstrated by documentation, while trust is established by the consistent behavior of systems and institutions over time.

The structural forces that have produced the current gaps — collective action problems, disclosure aversion, sovereignty concerns, first-mover disadvantages in standards — are not self-resolving. They will not be addressed by the passage of more rules, by the publication of more principles, or by the accumulation of more voluntary commitments. They require deliberate institutional design: the creation of bodies with the mandate and capacity to develop and maintain trust infrastructure; the negotiation of agreements that allocate liability and define mutual obligations; the investment of political capital in the unglamorous but essential work of cross-reference mapping, standard-setting, and confidence-building.

The four recommendations in Section 7 are directed at this work. They are not comprehensive — a complete agenda for AI trust infrastructure would be substantially longer. But they address root causes rather than symptoms, and they are achievable within the timeframes specified if the actors with authority to act on them choose to do so. We are not calling for a new global

institution. We are calling for the existing institutions to do the connective work that their current mandates permit but have not yet produced.

The window for building this infrastructure is not permanently open. As AI systems become more deeply embedded in critical infrastructure — in healthcare, finance, public administration, and the information environment — the costs of trust failures will increase, and the institutional capacity to redesign the underlying infrastructure will decrease. The patterns established now — in attestation practices, in audit trail formats, in incident reporting architectures — will shape the governance landscape for years. The unglamorous foundation is easier to lay before the building is occupied.

The "unglamorous foundation" this paper describes — attestation infrastructure, verification mechanisms, shared interpretive frameworks, accountability channels — is not, as some might assume, the domain exclusively of large institutions with dedicated compliance infrastructure. HHI's own governance ecosystem demonstrates that a small research institution can build and maintain operational trust infrastructure using open tooling and structured artifact discipline. The governance evidence trail produced by HHI's repositories — provenance manifests, checksum-bound audit logs, replay-capable telemetry, escalation persistence records, exportable audit bundles — constitutes, in miniature, the kind of trust substrate this paper argues the field needs. Trust infrastructure, as this paper argues, is a prerequisite. HHI's experience confirms that the prerequisite is achievable. The barrier is not technical. It is the collective decision to treat it as mandatory.

References

1. Baier, A. (1986). Trust and antitrust. *Ethics*, 96(2), 231–260.
2. Pettit, P. (1995). The cunning of trust. *Philosophy and Public Affairs*, 24(3), 202–225.
3. European Parliament and Council of the European Union. (2024). *Regulation (EU) 2024/1689 of the European Parliament and of the Council on Artificial Intelligence (AI Act)*. Official Journal of the European Union.
4. National Institute of Standards and Technology. (2023). *AI Risk Management Framework (AI RMF 1.0)*. NIST AI 100-1. U.S. Department of Commerce.
5. International Organization for Standardization. (2023). *ISO/IEC 42001:2023 — Information technology — Artificial intelligence — Management system*. ISO.
6. G7. (2023). *Hiroshima Process International Guiding Principles for Advanced AI Systems*. G7 Hiroshima AI Process.
7. Organisation for Economic Co-operation and Development. (2019, updated 2024). *Recommendation of the Council on Artificial Intelligence*. OECD/LEGAL/0449.

8. Council of Europe. (2024). *Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law* (CETS No. 225). Council of Europe Treaty Series.
9. International Civil Aviation Organization. (2016). *Safety Management Manual* (Doc 9859), 4th ed. ICAO.
10. International Civil Aviation Organization. (1944, amended). *Convention on International Civil Aviation* (Chicago Convention), Annex 13: Aircraft Accident and Incident Investigation. ICAO.
11. Basel Committee on Banking Supervision. (2017). *Basel III: Finalising Post-Crisis Reforms*. Bank for International Settlements.
12. European Commission. (2017). *EU-US Mutual Recognition Agreement for Pharmaceutical Inspections*. Official Journal of the European Union.
13. Financial Industry Regulatory Authority. (2023). *Report on Artificial Intelligence in the Securities Industry*. FINRA.
14. Global Legal Entity Identifier Foundation. (2012, updated). *LEI System Overview and Architecture*. GLEIF.
15. Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I. D., & Gebru, T. (2019). Model cards for model reporting. *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 220–229.
16. Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé III, H., & Crawford, K. (2018). Datasheets for datasets. *arXiv preprint arXiv:1803.09010*. Published in *Communications of the ACM*, 64(12), 86–92 (2021).
17. Raji, I. D., Smart, A., White, R. N., Mitchell, M., Gebru, T., Hutchinson, B., Smith-Loud, J., Theron, D., & Barnes, P. (2020). Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, 33–44.
18. Metcalf, J., Moss, E., & Boyd, D. (2019). Owning ethics: Corporate logics, Silicon Valley, and the institutionalization of ethics. *Social Research: An International Quarterly*, 86(2), 449–476.
19. Cihon, P., Maas, M. M., & Floridi, L. (2020). Should artificial intelligence governance be centralised? Design lessons from history. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*.
20. Dafoe, A. (2018). *AI Governance: A Research Agenda*. Future of Humanity Institute, University of Oxford.
21. Marchetti, R. (2021). *The governance of artificial intelligence: A global challenge*. *International Studies Review*, 23(4), 1926–1931.
22. Hadfield, G. K., & Clark, J. (2023). *Regulatory Markets: The Future of AI Governance*. MIT Press.

23. Ho, L., Barnhart, J., Trager, R., Bengio, Y., & Dafoe, A. (2023). International institutions for advanced AI. *arXiv preprint arXiv:2307.04699*.
 24. Krakovna, V., Uesato, J., Mikulik, V., Martic, M., Tomasev, N., Stepleton, T., Everitt, T., Zvyagelskaya, A., Jiang, R., & Leike, J. (2020). Avoiding side effects in complex environments. *Advances in Neural Information Processing Systems*, 33.
 25. Wachter, S., Mittelstadt, B., & Russell, C. (2021). Why fairness cannot be automated: Bridging the gap between EU non-discrimination law and AI. *Computer Law & Security Review*, 41, 105567.
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